

The Language of Business AnalyticsFoundational Concepts of Analytics(1) Business AnalyticsSource: Dataiku

The iterative process of exploring and analyzing business data to drive strategic planning and operations. Unlike Data Analytics, which focuses on the technical processing of data, Business Analytics focuses on applying those insights to solve broader organizational problems.

Example:

A retailer analyses its sales data to determine which products should be stocked more heavily.

(2) Data AnalyticsSource: Dataiku

The technical science of examining raw datasets to extract actionable conclusions and identify patterns. It is narrower than Business Analytics, focusing heavily on the mathematical and computational techniques required to clean, transform and model data.

Example:

A data analyst writing scripts in Python or R to clean raw census and structure it for statistical analysis.

(3) Business Intelligence (BI)Source: Gartner

The strategies and technologies used by enterprises to analyze current and historical business information. While analytics generally looks forward to predict or prescribe, BI looks backward to report on exactly what has already happened using structured dashboards.

Example: Using a reporting dashboard to track the exact monthly volume of citizen queries received by a municipality.

(4) Data-Driven Decision Making Source: Harvard Business School Online  
A management philosophy where strategic decisions are based on data analysis and interpretation rather than intuition, observation or hierarchy. It requires organizational trust in data accuracy and the willingness to let statistical guide operations.

Example:

A provincial government choosing where to build a new public clinic based strictly on demographic population density data rather than political preference.

(5) Big Data Source: The Decision Lab  
Extremely large, complex datasets that grow exponentially and cannot be managed by traditional data processing software. It is characterized by 3 V's: high volume, high velocity and high variety.

Example:

Processing the massive, continuous stream of unstructured social media text and structured traffic data to monitor public sentiment and city congestion in real-time.

(6) Descriptive Analytics Source: Gartner

The simplest stage of data analysis that summarizes historical data to understand what has happened in a given period. It forms the foundation of all reporting by organizing raw data into a digestible surface level format.

Example: Generating a monthly report in Excel that shows the total number of new infrastructure project applications received last year.

### (7) Diagnostic Analytics Source: Gartner

The analytical process of investigating descriptive data further to determine the root causes behind past outcomes. It answers the question, why did it happen, by identifying anomalies, discovering correlations and finding causal relationships.

#### Example:

Investing a sudden drop in internship applications by analyzing web ~~off~~ traffic data to discover a broken submission link on the online portal.

### (8) Predictive Analytics Source: Gartner

The use of historical data, statistical algorithms and machine learning techniques to forecast future events or trends. It shifts the focus from the past to answering, what is likely to happen, based on statistical probabilities.

#### Example:

Building a Logistic Regression or Random Forest model to forecast the probability of power grid failure during winter months based on historical data.

### (9) Prescriptive Analytics Source: Gartner

The most advanced form of analytics that ~~also~~ recommends specific actions to achieve desired outcomes. It answers, what should we do about it, by simulating various scenarios and suggesting the optimal path forward.

Example: An algorithm that not only predicts heavy traffic congestion but also automatically reroutes logistics and delivery trucks to the fastest alternative routes.

#### (10) Data Literacy

Source: Edozi

The ability to read, understand, create and communicate data as information. It is a critical skill that empowers employees at all levels to ask the right questions of data and interpret charts and metrics accurately.

Example:

Training non-technical policy advisers to interpret SQL query results and statistical graphs so they can make informed legislative decisions.

### Data, Measurement & Reporting

#### (11) Structured Data

Source: IBM

• Structured Data follows a strict, predefined format like rows and columns. It uses a rigid layout like tables or spreadsheets, making it easy to search and analyse.

Example:

A bank stores customer names, account numbers and balances in database tables.

#### (12) Unstructured Data

Source: IBM

Unstructured Data has no fixed shape or exist in raw native formats. This comes as free form text, audio, video or image.

Example

A company analyses customer reviews on social media to

understand how customers feel about its products.

### (13) Data Warehouse

Source: Amazon Web Services

A central storage that stores cleaned, organized historical data from different sources for reporting and analysis. It is designed to make large amount of business data easier to query and use for decision-making.

Example:

(iii) A retailer stores sales, inventory and customer data in one data warehouse so analysts can pull the data and study & assess how the business is performing.

### (14) Data Mining.

Source: IBM

The process of examining large datasets to discover insightful patterns, relationships or trends. This process utilizes statistical methods and machine learning to find information that may not be obvious.

Example:

An online store uses data mining to ~~select~~ identify products that customers regularly buy together.

### (15) Data Visualization.

Source: IBM

This is to present information using charts, graphs, plots or other visual forms. It is to help people identify patterns, trend & relationships more easily than looking at raw number alone.

Example:

A company uses a line graph to show how monthly sales have changed over the past year.

#### (16) Dashboard

Source: Tableau

A dashboard brings together important information and visualisations in one place so that performance can be monitored easily. Dashboards often display KPIs, charts and business measures on a single screen.

##### Example:

A sales manager uses a Power BI dashboard to monitor sales, revenue and customer performance.

#### (17) Metric

Source: IBM

A metric is a measurable value used to track or describe some aspect of business activity or performance. Metrics can be simple measurements such as sales, response time or number of orders.

##### Example:

A customer service department measures the average time taken to respond to customer enquiries.

#### (18) Key Performance Indicator (KPI)

Source: IBM

A KPI is a specific measurable value that is important to achieving a business objective. Unlike a general metric, a KPI is selected because it helps the business monitor a critical area of performance, often against a target.

##### Example:

A call center may use average customer response time as a KPI, with a target of responding within one minute.

## (10) Benchmark

Source: IBM

A benchmark is a reference point or standard used to compare actual performance. It helps a business determine whether its results are meeting an expected level or falling below it.

### Example:

A company compares its customer satisfaction score with an industry benchmark to see how its performance compares.

## (20) Segmentation

Source: HubSpot

This is the process of dividing customers or data into smaller groups based on shared characteristics. Businesses use segmentation to understand different groups and tailor products, services or marketing to them.

### Example:

An online retailer divides customers into groups based on age, location and purchasing behaviour and send different promotions to each group.

## Customer & Sales KPIs

### (21) Conversion Rate

Source: Google Ads Help

A percentage of people who complete a desired action out of the total number of people who had the opportunity to do so. The action could be making a purchase, signing up for a service or submitting a form.

### Example:

An online store has 1000 visitors and 50 of them makes purchase, giving the store a 5% conversion rate.

(22) Customer Acquisition Cost (CAC) Source: Shopify  
CAC is the average amount a business spends to gain one new customer. It includes costs related to activities such as marketing and sales.

Example:

A company spends R20,000 on marketing & gains 200 new customers, giving it a CAC of R100 per customer.

(23) Customer Lifetime Value (CLV / LTV) Source: Shopify

Customer Lifetime Value estimates the total value a business can receive from a customer over the entire relationship with that customer. It considers factors such as how much the customer spends, how often they purchase and how long they remain a customer.

Example:

A clothing customer spends R500 per purchase, buys four times a year and remains a customer for three years, creating a substantial lifetime value for the business.

(24) Churn Rate Source: Stripe

Churn rate is the percentage of customers who stop using a company's product or service during a specific period. It is particularly important for businesses that depend on subscriptions or repeat customers.

Example:

A streaming service starts the month with 1000 subscribers and loses 50, resulting in a 5% churn rate.

### (25) Customer Retention Rate

Source: Shopify

Customer retention rate measures the percentage of existing customers that a business manages to keep over a specific period. It shows how successfully a company maintains its customer base.

#### Example:

A company starts with 1000 customers and keeps 900 of its existing customers during the year, giving it a strong retention rate.

### (26) Net Promoter Score (NPS)

Source: Bain & Company

This measures customer loyalty based on how likely customers are to recommend a company's product or service to others. Customers are normally asked to give a rating from 0 to 10, which is then used to calculate the NPS.

#### Example:

A bank asks its customers how likely they are to recommend the bank to a friend or family member & use the responses to monitor customer loyalty.

### (27) Average Order Value (AOV)

Source: Shopify

AOV measures the average amount of money spent by customers in the single transaction. It is commonly calculated by dividing total revenue by number of orders.

#### Example:

If an online store makes ₹50,000 from 1000 orders, its AOV is 50.

## (28) Sales Funnel

Source: HubSpot

A sales funnel shows the stages potential customers move through before becoming paying customers. The number of potential customers usually becomes smaller as they move from initial awareness through to the final purchase.

### Example:

An online business may track customers from seeing an advertisement, visiting its website, adding a product to a basket and finally completing the purchase.

## Marketing and Campaign Analytics.

## (29) Marketing Campaign

Source: Hubspot

A marketing campaign is a planned and coordinated set of marketing activities designed to achieve a specific business goal. It can use several channels, such as social media, email, websites or advertising.

### Example:

A clothing company launches a campaign on Instagram, email and its website to promote a new winter collection.

## (30) Post - Campaign Analysis

Source: Hubspot

Post - Campaign Analysis is the process of reviewing campaign results after the campaign has run. It helps a business understand what worked, what did not work and what could be improved next time.

### Example:

After an online advertising campaign, a company reviews sales, clicks, ~~even~~ conversions and spending to assess its results.

### (81) Incremental Revenue

Source: Google Ads Help

Incremental Revenue is the additional revenue generated because of a specific marketing activity or campaign. It focuses on the revenue that would not have been generated without that campaign.

#### Example:

If customers exposed to a campaign generate R100,000 while a comparable control group generates R70,000 the incremental revenue is R30,000.

### (82) Return on Investment (ROI)

Source: Google Ads Help

ROI measures the financial return generated in relation to the cost of an investment. In marketing, it can be used to determine whether the profit generated by a campaign justified the money spent.

#### Example:

A company spends R10,000 on a campaign and generates R15,000 in profit resulting in a 50% ROI.

### (83) Return on Ad Spend (ROAS)

Source: Google Ads Help

ROAS measures how much revenue is generated for the money spent on advertising. It is calculated by dividing converting value or revenue by advertising spend.

#### Example:

If a business spends R5,000 on advertising and generates R20,000 in revenue, its ROAS is 4 or 400%.

(34) Click-Through Rate (CTR) Source: Google Ads Help

CTR is the percentage of people who click on an advertisement or link after seeing it. It is calculated by dividing the number of clicks by the number of impressions.

Example:

An advert receives 50 clicks from 1000 impressions giving it a CTR of 5%.

(35) Bounce Rate Source: Google Analytics Help

This measures the percentage of website or app sessions that were not considered engaged. It can be used to understand whether visitors are interacting meaningfully with digital content.

Example:

A website with many visitors leaving without meaningful engagement may have a high bounce rate.

(36) A/B Testing Source: Google Analytics Help

A/B Testing compares two or more versions of a webpage or marketing element by showing the versions to different users. The results are then compared to determine which version performs better against a chosen goal.

Example:

An online retailer tests two different versions of a product page to see which one produces more purchases.

~~Example:~~

(37) Lead Generation.

Source: Google Analytics Help

Lead generation is the process of attracting potential customers and capturing their information so that a business can

develop the relationship and potentially convert them into customers. Common methods include forms, free trials, webinars and newsletters.

#### Example:

A software company offers a free guide in exchange for a visitor's email address.

#### (38) Marketing Attribution

Source: HubSpot

Marketing Attribution is the process of identifying and assigning credit to marketing channels or customer touchpoints that contributed to a conversion. It helps businesses understand how different marketing activities contribute to results.

#### Example:

A company tracks whether a customer first came from Google, later interacted with an email and finally purchased after clicking a social media advert.